

FREE VIBRATIONS OF RECTANGULAR CURVED PANELS

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Summary—The exact solution and some approximate solutions for the free vibrations of uniform rectangular curved panels with simply supported edges are given. Exact solutions are not available for all boundary conditions and a general energy method, in which the mid-surface displacements are represented by truncated double-power series in the mid-surface co-ordinates of the panel, is outlined. Numerical results for the lowest natural frequency of shallow clamped panels with five different side-ratios and with curvature and thickness within a wide range are given. The range of panel geometry, for which an approximate frequency equation for clamped panels gives accurate results, is defined.

NOTATION

a	radius of curvature
b	curved length of panel
k	$(t/a)^2/12$
l	length of panel in x direction
m, n	integers
$\{q\}$	column vector of power series coefficients
$\{\bar{q}\}$	column vector of amplitudes of $\{q\}$
t	thickness
u, v, w	displacements in x, y and z directions
x, y, z	co-ordinates
$\{x\}, \{x_2\}$	column vectors of power series coefficients
A, B, C	amplitudes of u, v and w displacements
$[B]$	transformation matrix relating $\{x\}$ to $\{q\}$
$[B]^T$	transpose of $[B]$
C_1, C_2, C_3	constants in equations (3) and (4)
$[C], [C_1], [C_2]$	matrices
E	Young's modulus
J	number of constraints
K_1, K_2, K_0	constants in equation (2)
$[K]$	matrix in strain energy expression
M, N	numbers of terms in power series in x and y
$[M]$	matrix in kinetic energy expression
T	kinetic energy
$U(x), V(x), W(x)$	functions of x defining variation of u, v and w in x direction
U_{mn}, V_{mn}, W_{mn}	coefficients of terms in power series displacement functions
V	strain energy
λ	$m\pi a/l$, m = no. of half-waves in x direction
μ	$n\pi a/b$, n = no. of half-waves in y direction
ν	Poisson's ratio
ρ	density
ω	circular frequency
Δ	$\rho\omega^2(1-\nu^2)a^2/E$
Ω	$\rho\omega^2(1-\nu^2)l^2b^2/Et^2$
$q^{(i)}$	i th element in column vector $\{q\}$

INTRODUCTION

AT THE present time there are available many computer programs for determining the natural frequencies of a wide range of elastic structures. Even so, it is useful to be able to refer to data from which the natural frequencies of particular structures may be calculated easily. Usually this is possible only for simple structures such as beams, plates, etc. Sometimes assumptions in the analysis allow simple equations to be derived for the natural frequencies. These assumptions often restrict the range of validity of the resulting equations and it is important to establish the region over which the simplified equations are valid.

Recently, Sewall¹ has evaluated some equations for the natural frequencies of shallow curved rectangular panels with all the edges either simply supported or clamped. The range of curvature and thickness for which these equations give accurate results has not been completely defined in ref. 1. The object of this paper is to define the region over which Sewall's equations are valid and to give some data which extend the range of panel geometries for which the natural frequencies may be obtained.

ANALYSIS

Fig. 1 shows the type of panel considered here. The panel is a piece of a cylindrical shell bounded by four planes, two perpendicular to the axis of the cylinder at $x = 0$ and

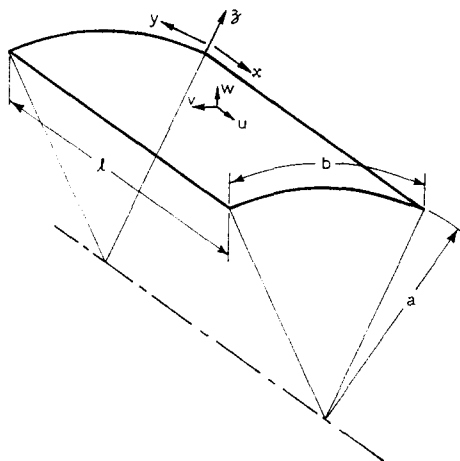


FIG. 1. Rectangular curved panel.

$x = l$, and two which contain the axis of the cylinder and one of the generators, either at $y = 0$ or $y = b$. The co-ordinate system used in the analysis is also shown in Fig. 1.

Simply supported edges

The conditions at a simply supported edge are

Displacement normal to mid-surface	= 0
Displacement along edge	= 0
Direct force normal to edge	= 0
Edge bending moment	= 0

The equations of motion for an element of a cylindrical shell are given by Flügge². Exact solutions of these equations can be obtained when the two straight edges, at $y = 0$ and $y = b$, are simply supported and for any specified boundary conditions along the curved edges, at $x = 0$ and $x = l$, by the methods used by Forsberg³ and Warburton⁴ for complete cylindrical shells. When the boundary conditions are uniform along the curved edges the solutions are of the form

$$\begin{aligned}u &= U(x) \sin \frac{n\pi y}{b} \cos \omega t \\v &= V(x) \cos \frac{n\pi y}{b} \cos \omega t \\w &= W(x) \sin \frac{n\pi y}{b} \cos \omega t\end{aligned}$$

If this solution is substituted into the equations of motion the resulting three simultaneous differential equations in U , V and W may be solved to give the natural frequencies and modes of free vibration.

When all four edges are simply supported the solution is of the form

$$\left. \begin{aligned}u &= A \cos \frac{m\pi x}{l} \sin \frac{n\pi y}{b} \cos \omega t \\v &= B \sin \frac{m\pi x}{l} \cos \frac{n\pi y}{b} \cos \omega t \\w &= C \sin \frac{m\pi x}{l} \sin \frac{n\pi y}{b} \cos \omega t\end{aligned} \right\} \quad (1)$$

Substituting this solution into the equations of motion gives the following frequency equation

$$\Delta^3 - K_2 \Delta^2 + K_1 \Delta - K_0 = 0 \quad (2)$$

where, if terms containing k^2 are neglected,

$$\begin{aligned}K_2 &= 1 + \left(\frac{3-\nu}{2}\right) (\lambda^2 + \mu^2) + k \left[1 + (\lambda^2 + \mu^2)^2 + 3 \left(\frac{1-\nu}{2}\right) \lambda^2 - \left(\frac{3+\nu}{2}\right) \mu^2 \right] \\K_1 &= \left(\frac{1-\nu}{2}\right) [(\lambda^2 + \mu^2)^2 + (3+2\nu) \lambda^2 + \mu^2] \\&\quad + k \left[\left(\frac{3-\nu}{2}\right) (\lambda^2 + \mu^2)^3 - \left(\frac{5-\nu}{2}\right) (\lambda^2 + \mu^2)^2 + 4(1-\nu) \lambda^4 + (3-2\nu) \lambda^2 + (2-\nu) \mu^2 \right] \\K_0 &= \frac{1}{2}(1+\nu) (1-\nu)^2 \lambda^4 + k \left(\frac{1-\nu}{2}\right) [(\lambda^2 + \mu^2)^4 - 2\nu \lambda^6 - 6\lambda^4 \mu^2 - 2(4-\nu) \lambda^2 \mu^4 \\&\quad - 2\mu^6 + (4-3\nu^2) \lambda^4 + 2(2-\nu) \lambda^2 \mu^2 + \mu^4]\end{aligned}$$

The relative amplitudes of the three displacements given by equation (1) are

$$\frac{A}{C} = \frac{C_1}{C_2} \quad \text{and} \quad \frac{B}{C} = \frac{C_3}{C_2} \quad (3)$$

where

$$\begin{aligned}C_1 &= \left(\frac{1-\nu}{2}\right) \{\nu \lambda^3 - \lambda \mu^2 + k[\lambda^5 - \lambda \mu^4 + 3\nu \lambda^3]\} - \Delta \left\{ \nu \lambda + k \left[\lambda^3 - \left(\frac{1-\nu}{2}\right) \lambda \mu^2 \right] \right\} \\C_2 &= \Delta^2 - \Delta \left[\left(\frac{3-\nu}{2}\right) (\lambda^2 + \mu^2) + k \left(\frac{1-\nu}{2}\right) (3\lambda^2 + \mu^2) \right] \\&\quad + \left(\frac{1-\nu}{2}\right) \{(\lambda^2 + \mu^2)^2 + k[3\lambda^4 + \mu^4 + 2(1-\nu) \lambda^2 \mu^2]\} \\C_3 &= \left(\frac{1-\nu}{2}\right) \{(2+\nu) \lambda^2 \mu + \mu^3 + k[2\lambda^4 \mu + 2\lambda^2 \mu^3 + \mu^3]\} - \Delta \left[\mu + k \left(\frac{3-\nu}{2}\right) \lambda^3 \mu \right]\end{aligned}$$

For shallow shells, the in-plane inertia is small and the in-plane inertia terms may be omitted from the equations of motion. The exact solution of these simplified equations still gives the displacements in the form of equation (1). The resulting frequency equation and constants in the relative amplitude relations (3) are

$$\left. \begin{aligned} C_2 \Delta &= (1-\nu^2) \lambda^4 + k[(\lambda^2 + \mu^2)^4 - 2\nu\lambda^6 - 6\lambda^4 \mu^2 - 2(4-\nu) \lambda^2 \mu^4 \\ &\quad - 2\mu^6 + (4-3\nu^2) \lambda^4 + 2(2-\nu) \lambda^2 \mu^2 + \mu^4] \\ C_1 &= \nu\lambda^3 - \lambda\mu^2 + k(\lambda^5 - \lambda\mu^4 + 3\nu\lambda^3) \\ C_2 &= (\lambda^2 + \mu^2)^2 + k[3\lambda^4 + 2(1-\nu) \lambda^2 \mu^2 + \mu^4] \\ C_3 &= (2+\nu) \lambda^2 \mu + \mu^3 + k(2\lambda^4 \mu + 2\lambda^2 \mu^3 + \mu^3) \end{aligned} \right\} \quad (4)$$

For thin shells Donnell's approximation may be used, i.e. z , the co-ordinate through the shell thickness is neglected compared with the radius of curvature a , in the expressions for strains and stress resultants. The equations of motion resulting from this approximation are given by Flügge⁵ and again neglecting in-plane inertia we have

$$\left. \begin{aligned} \Delta &= \frac{(1-\nu^2) \lambda^4}{(\lambda^2 + \mu^2)^2} + k(\lambda^2 + \mu^2)^2 \\ \frac{A}{C} &= \frac{\lambda(\nu\lambda^2 - \mu^2)}{(\lambda^2 + \mu^2)^2} \\ \frac{B}{C} &= \frac{\mu[\mu^2 + (2+\nu) \lambda^2]}{(\lambda^2 + \mu^2)^2} \end{aligned} \right\} \quad (5)$$

The first of equations (5) is the same as that derived by Sewall¹ for simply supported panels using a Rayleigh-Ritz procedure. Beam modes were used for the displacement expressions and in this case they are the exact solutions (1).

Any homogeneous boundary conditions

Exact solutions for the free vibration problem for rectangular curved panels are only known when the two straight edges are simply supported. For other boundary conditions there are a wide variety of methods which would lead to approximate solutions of adequate accuracy. It is often difficult at the outset to judge which of these methods is the most convenient to apply to a particular problem. In this case the author has chosen a method with which he is familiar and for which there are standard computer programs available for solving the numerical equations.

The method is essentially the same as the Rayleigh-Ritz method with one significant difference. In the Rayleigh-Ritz method the assumed functions describing the displacements of the structure each satisfy the geometric boundary conditions. In the method used here, the mid-surface displacements are assumed in the form of power series in the x and y co-ordinates. The solution is obtained by finding stationary values of the Hamiltonian

$$\int_1^2 (T - V) dt$$

subject to the constraints required to satisfy the geometric boundary conditions.

Assuming that the mid-surface displacements may be expressed in the form

$$\left. \begin{aligned} u &= \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} U_{mn}(x/l)^m (y/b)^n \\ v &= \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} V_{mn}(x/l)^m (y/b)^n \\ w &= \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} W_{mn}(x/l)^m (y/b)^n \end{aligned} \right\} \quad (6)$$

An expression for the strain energy of the panel may be derived from the one given for a cylinder in appendix 2 of ref. 4. The positive direction for the normal displacement in

ref. 4 is different from that used here and a number of sign changes are required in this expression.

After substituting for the displacements (6) and integrating over the area of the panel, we have the strain energy in the form

$$V = \frac{1}{2} [x] [K] \{x\} \quad (7)$$

where $\{x\}$ is a column of the $3MN$ coefficients U_{mn} , V_{mn} and W_{mn} and $[K]$ is a square matrix.

The kinetic energy, neglecting rotatory inertia is

$$T = \int_0^l \int_0^b \int_{-t/2}^{+t/2} \frac{1}{2} \rho (\dot{u}^2 + \dot{v}^2 + \dot{w}^2) (1 + z/a) dz dy dx$$

Substituting for the displacements and integrating gives

$$T = \frac{1}{2} [\dot{x}] [M] \{\dot{x}\} \quad (8)$$

The displacement expressions (6) do not satisfy the geometric boundary conditions, except for a panel with all the edges free. For other boundary conditions, linear expressions in the coefficients U_{mn} , V_{mn} and W_{mn} must be satisfied. The following two examples illustrate this.

(a) $u = 0$ at $x = 0$

Therefore

$$\sum_{m=0}^{M-1} \sum_{n=0}^{N-1} U_{mn} \left(\frac{x}{l}\right)^m \left(\frac{y}{b}\right)^n = 0 \quad \text{at } x = 0$$

Therefore

$$\sum_{n=0}^{N-1} U_{0n} \left(\frac{y}{b}\right)^n = 0$$

This must be satisfied for all y , $0 \leq y \leq b$ and thus the coefficient of each y term must vanish. Therefore

$$U_{0n} = 0, \quad n = 0, 1, 2, \dots, N-1$$

(b) $\partial w / \partial y = 0$ at $y = b$

Therefore

$$\sum_{m=0}^{M-1} \sum_{n=0}^{N-1} W_{mn} \frac{n}{b} \left(\frac{x}{l}\right)^m \left(\frac{y}{b}\right)^{n-1} = 0 \quad \text{at } y = b$$

Therefore

$$\sum_{m=0}^{M-1} \sum_{n=0}^{N-1} W_{mn} \frac{n}{b} \left(\frac{x}{l}\right)^m = 0$$

This must be satisfied for all x , $0 \leq x \leq l$ and thus the coefficient of each x term must vanish. Therefore

$$\sum_{n=0}^{N-1} n W_{mn} = 0, \quad m = 0, 1, 2, \dots, M-1$$

Each boundary condition at $x = 0$ or $x = l$ results in N constraints on the coefficients and each boundary condition at $y = 0$ or $y = b$ results in M constraints.

A solution of the free vibration problem may be obtained by applying Hamilton's principle, i.e. by finding the stationary values of

$$\int_{t_1}^{t_2} (T - V) dt$$

subject to the constraints required to satisfy the boundary conditions. This is a standard type of problem dealt with in most texts on the calculus of variations. It requires the introduction of Lagrange multipliers, one for each constraint.

The size of the system of equations resulting from this method is $3MN$ plus the number of constraints. This may be large when M and N are large enough to give adequate accuracy. An alternative method which reduces the size of the matrices but is in effect identical with the Lagrange multiplier method is given below.

The constraints may be written in the form

$$[C]\{x\} = 0 \quad (9)$$

If there are J constraints, $[C]$ is a $J \times 3MN$ matrix. These equations are used to eliminate J of the coefficients $\{x\}$ in terms of the remaining $3MN - J$ coefficients.

Partitioning (9) we have

$$[C_1]\{q\} + [C_2]\{x_2\} = 0$$

where $\{x_2\}$ is a column of the J coefficients to be eliminated and $\{q\}$ is a column of the remaining $3MN - J$ coefficients. Therefore

$$\{x_2\} = -[C_2^{-1}][C_1]\{q\} \quad (10)$$

From this we may now form a transformation matrix $[B]$ where

$$\{x\} = [B]\{q\}$$

This transformation is formed from the J equations (10) and the $3MN - J$ relations

$$x^{(i)} = q^{(i)}, \quad i = 1, 2, \dots, 3MN - J$$

The elements of $\{q\}$ are in effect generalized co-ordinates of the system. The strain energy and kinetic energy may be obtained in terms of $\{q\}$ by substituting for $\{x\}$ in equations (7) and (8).

$$V = \frac{1}{2}[q][B]^T[K][B]\{q\}$$

$$T = \frac{1}{2}[\dot{q}][B]^T[M][B]\{\dot{q}\}$$

Application of Hamilton's principle, which gives the Lagrange equations, or the Rayleigh-Ritz method gives

$$[B]^T\{[K] - \omega^2[M]\}[B]\{\bar{q}\} = 0 \quad (11)$$

This system of $3MN - J$ equations constitutes an algebraic eigenvalue problem and standard computer programs are available for its solution.

All edges clamped

The panel is symmetric about $x = l/2$ and $y = b/2$. It is more efficient to solve four separate problems for one-quarter of the panel, with the necessary symmetric or anti-symmetric boundary conditions along the lines of symmetry, than to solve for all the modes in the complete panel at once.

For all four cases the edges at $x = 0$ and $y = 0$ are the outside edges of the panel and the geometric boundary conditions are

At $x = 0$

$$u = v = w = \frac{\partial w}{\partial x} = 0$$

At $y = 0$

$$u = v = w = \frac{\partial w}{\partial y} - \frac{v}{a} = 0$$

The four cases are the four permutations of the boundary conditions along the lines of symmetry.

At $x = l/2$

$$u = \frac{\partial w}{\partial x} = 0 \quad \text{or} \quad v = w = 0$$

At $y = b/2$

$$v = \frac{\partial w}{\partial y} = 0 \quad \text{or} \quad u = w = 0$$

Sewall derived an approximate frequency equation for the modes of vibration of a clamped panel by assuming the displacements for a particular mode to be of the form

$$u = A \frac{d}{dx} [X_m(x)] Y_n(y)$$

$$v = BX_m(x) \frac{d}{dy} [Y_n(y)]$$

$$w = CX_m(x) Y_n(y)$$

$X_m(x)$ and $Y_n(y)$ are the m th and n th mode shapes for a clamped-clamped beam

$$X_m(x) = \cosh \beta_m \frac{x}{l} - \cos \beta_m \frac{x}{l} - \gamma_m \left(\sinh \beta_m \frac{x}{l} - \sin \beta_m \frac{x}{l} \right)$$

β_m is a solution of

$$\cosh \beta_m \cos \beta_m = 1$$

and

$$\gamma_m = \frac{\cosh \beta_m - \cos \beta_m}{\sinh \beta_m - \sin \beta_m}$$

Similar expressions for $Y_n(y)$ are obtained by replacing x and m by y and n .

The following frequency equation is obtained by the Rayleigh-Ritz method, making Donnell's approximation and neglecting in-plane inertia.

$$\frac{\rho(1-\nu^2)\omega^2}{E} = \frac{t^2}{12l^2 b^3} \left[\left(\frac{b}{l} \right)^2 \beta_m^4 + \left(\frac{l}{b} \right)^2 \beta_n^4 + 2\bar{\beta}_m \bar{\beta}_n \right] + \frac{\beta_m^4 \beta_n^4 \left(1 - \left(\frac{\nu \bar{\beta}_m}{\beta_m^2} \right)^2 - \left(\frac{\bar{\beta}_n}{\beta_n^2} \right)^2 + \left(\frac{1-\nu}{2} \right) \bar{\beta}_m \bar{\beta}_n \left[\frac{b^2}{l^2} (\beta_m^4 - \nu^2 \bar{\beta}_m^2) + \frac{l^2}{b^2} (\beta_n^4 - \bar{\beta}_n^2) \right] + (\nu \bar{\beta}_m \bar{\beta}_n)^2 \right)}{\alpha^2 \left[\beta_m^4 \beta_n^4 + \left(\frac{1-\nu}{2} \right) \bar{\beta}_m \bar{\beta}_n \left(\frac{b^2}{l^2} \beta_m^4 + \frac{l^2}{b^2} \beta_n^4 \right) - \nu (\bar{\beta}_m \bar{\beta}_n)^2 \right]} \quad (12)$$

where

$$\bar{\beta}_m = \gamma_m \beta_m (\gamma_m \beta_m - 2) \quad \text{and} \quad \bar{\beta}_n = \gamma_n \beta_n (\gamma_n \beta_n - 2)$$

NUMERICAL RESULTS FOR SHALLOW CLAMPED PANELS

Sewall's equation (12) gives the natural frequencies of rectangular curved panels with clamped edges in a convenient form. The quantities in this equation are readily available, see, for example, ref. 6. The range over which this equation is valid has been found by comparing the results obtained from it with those obtained from the solution of equation (11). The solutions of equation (11) are upper bound and converge monotonically to the exact solution of Flügge's equations of motion, as the number of terms in the power series representing the displacements is increased. The number of terms in the series may not be increased indefinitely because of restrictions on computer store and also because the equations may become ill-conditioned if large numbers of terms are used. Hence, the accuracy of the solutions has been assessed by comparing the convergence with increasing numbers of terms.

In Figs. 2-6 a frequency parameter, Ω , corresponding to the lowest natural frequency, is plotted against the geometric parameter bl/at for five different side-ratios. The notation (m, n) used on these figures indicates that the normal displacement, in the modes corresponding to these natural frequencies, has m and n half-waves in the x and y direction respectively.

It is estimated that the double-power series solutions are within a few per cent of the true solutions. For modes with a large number of half-waves in the y direction, the error may be as large as 10 per cent. For these modes, terms up to $x^5 y^8$ and $x^7 y^7$ were used to represent the displacements of one-quarter of the panels. The expression for the frequency parameter, Ω , derived from Sewall's equation (12), is independent of b/a for a given b/l and bl/at , but the power series solution is not. The results given in Figs. 2-6 were calculated

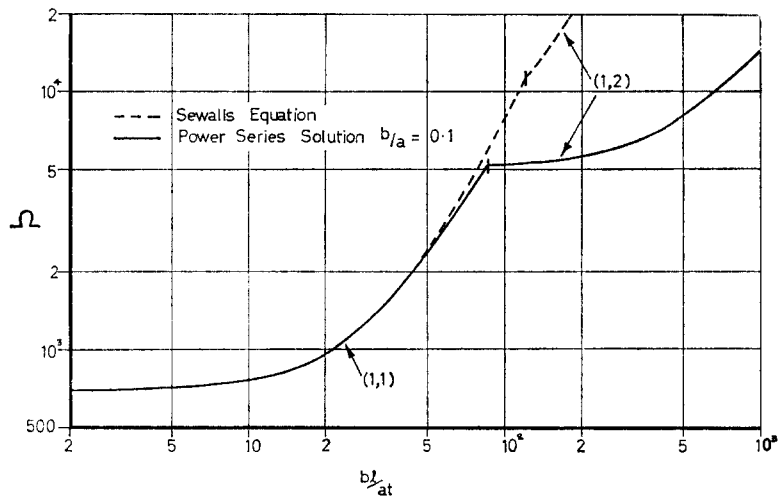


FIG. 2. Lowest natural frequency of clamped rectangular curved panels, $b/l = 0.25$.

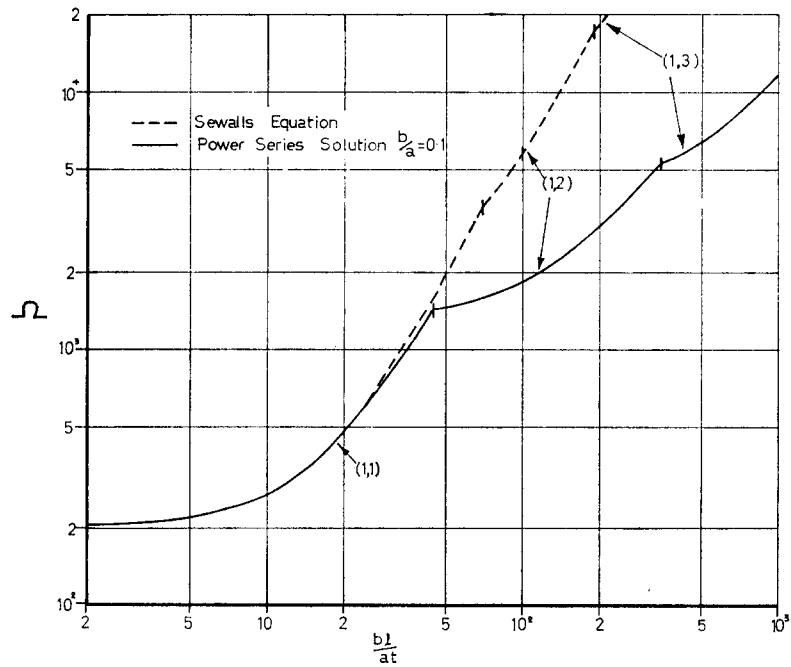


FIG. 3. Lowest natural frequency of clamped rectangular curved panels, $b/l = 0.5$.

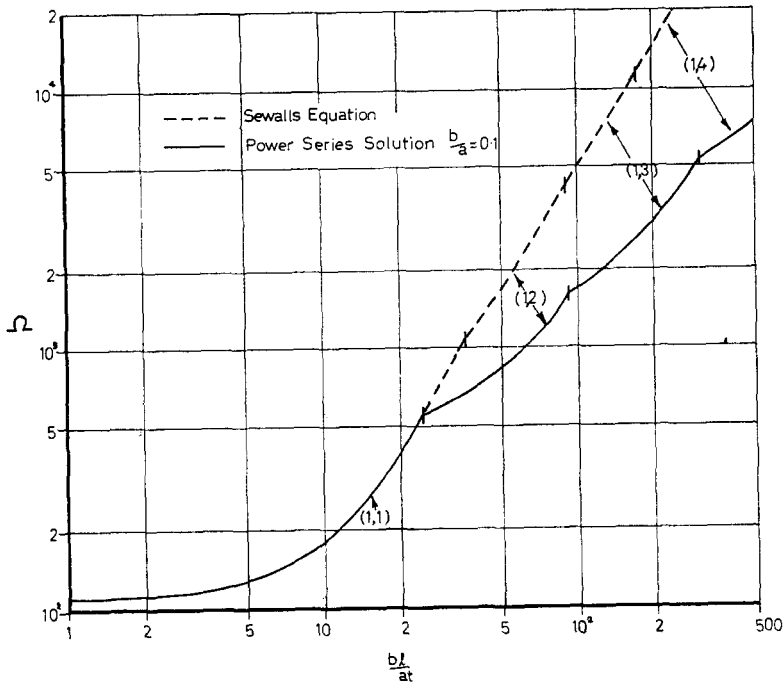


FIG. 4. Lowest natural frequency of clamped rectangular curved panels, $b/l = 1.0$.

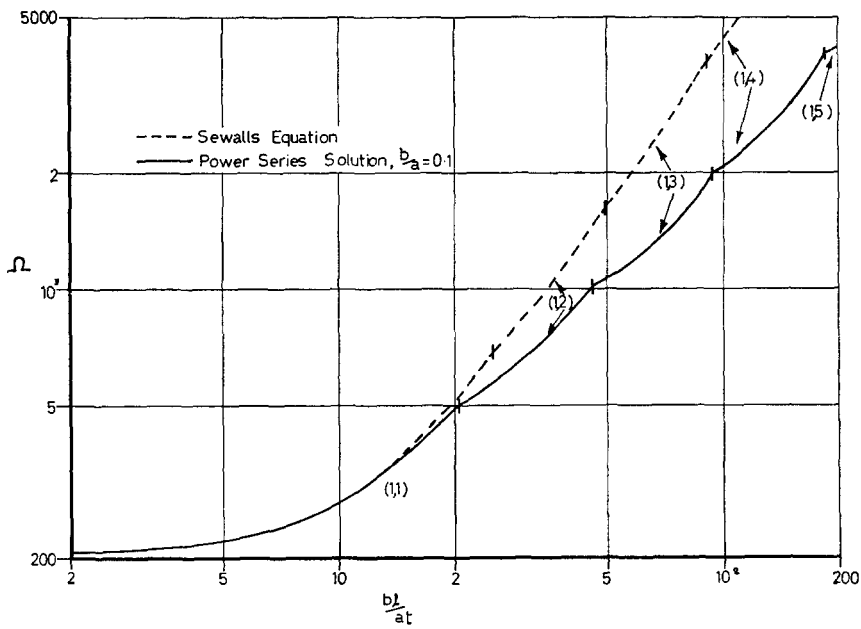


FIG. 5. Lowest natural frequency of clamped rectangular curved panels, $b/l = 2.0$.

for $b/a = 0.1$, but they may be used for shallow panels generally. At $b/a = 1.0$ the results from equation (11) would be about 2-3 per cent less than those for $b/a = 0.1$.

It may be seen from Figs. 2-6 that Sewall's equation gives accurate values for the lowest natural frequency for low values of the parameter bl/at . The value of bl/at at which Sewall's equation diverges from the more accurate results depends on the side-ratio, but it may be seen that if either bl/at or b^2/at is less than 20 then Sewall's equation gives accurate results.

Sewall's equation becomes inaccurate as bl/at increases because the assumed displacements used in its derivation do not represent very accurately the true displacements. In

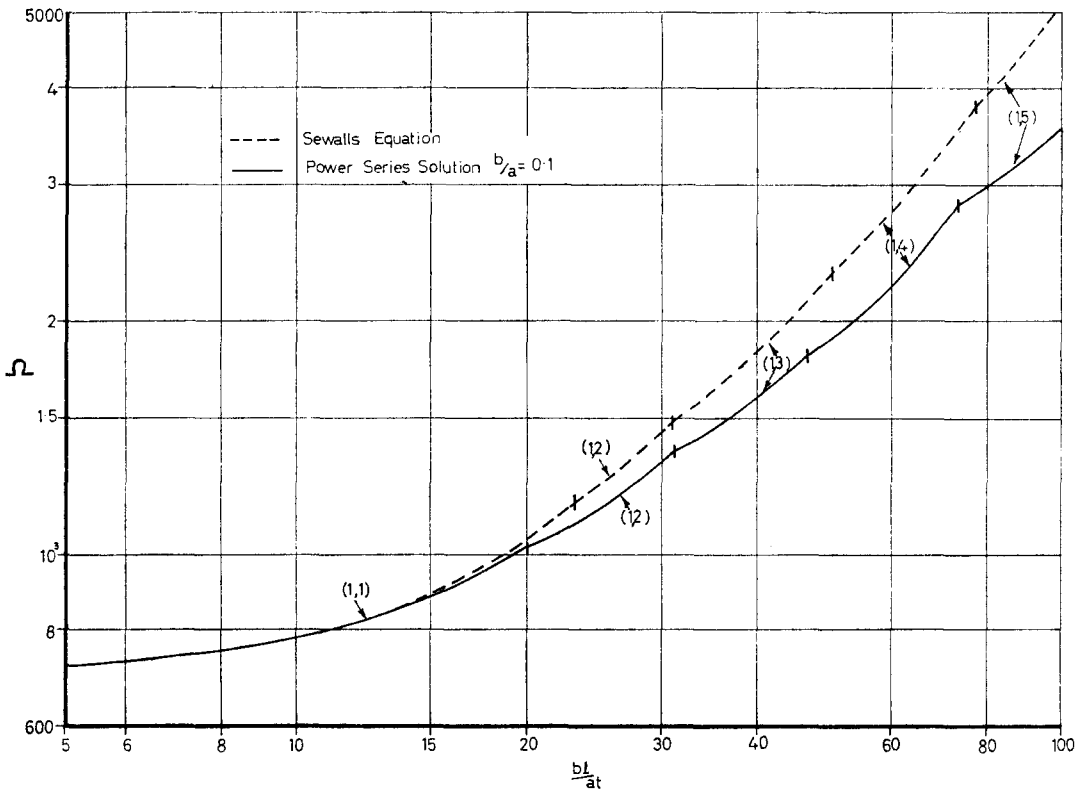


FIG. 6. Lowest natural frequency of clamped rectangular curved panels,
 $b/l = 4.0$.

particular, the representation of the variation of the v displacement in the y direction is poor for modes with more than one half-wave in the y direction. For thick panels with large curvatures, these misrepresentations have a negligible effect on the natural frequencies because the stretching strain energy is very small compared with the bending strain energy. For thinner, more highly curved panels, the stretching becomes more significant and the natural frequencies are affected.

Some of the normal modes of flat, clamped, square plates do not have nodal lines which are approximately parallel to the edges of the plate. These modes in effect replace modes which would have identical natural frequencies, because of symmetry, if the nodal lines were parallel to the edges. These modes are identified as $(m, n) \pm (n, m)$ modes because they may be approximated by proportions of two assumed modes, (m, n) and (n, m) , which do have nodal lines parallel to the edges. Warburton⁷ showed that the nodal patterns of these modes are very sensitive to the introduction of asymmetry by making the side-ratio slightly different from unity. A very similar effect occurs when asymmetry is

introduced by curving the plate in one direction. Fig. 7 shows the transformation of the mode shapes of the $(3, 1) \pm (1, 3)$ modes to $(1, 3)$ and $(3, 1)$ modes by giving the panel a constant curvature in one direction. For $bl/at = 8$ the rise of the square curved panel is approximately equal to the thickness. It may be seen that curvatures of this order change the nodal pattern considerably.

Most of the normal modes of a clamped rectangular panel have nodal lines which are approximately parallel to the edges of the panel. The order of the modes of a panel, i.e. which modes have the lowest natural frequencies etc., depends on the panel geometry. Hence it is expected that some panels have two modes with equal natural frequencies. If the two modes have different symmetry or anti-symmetry in either the x direction or the y direction, then two modes with equal natural frequencies occur. If the two modes have the same symmetry or anti-symmetry in the x direction and in the y direction, two modes with different natural frequencies occur. The mode shapes of these two modes may be rather complex and they may be made up, approximately, of combinations of the two simpler modes they replace.

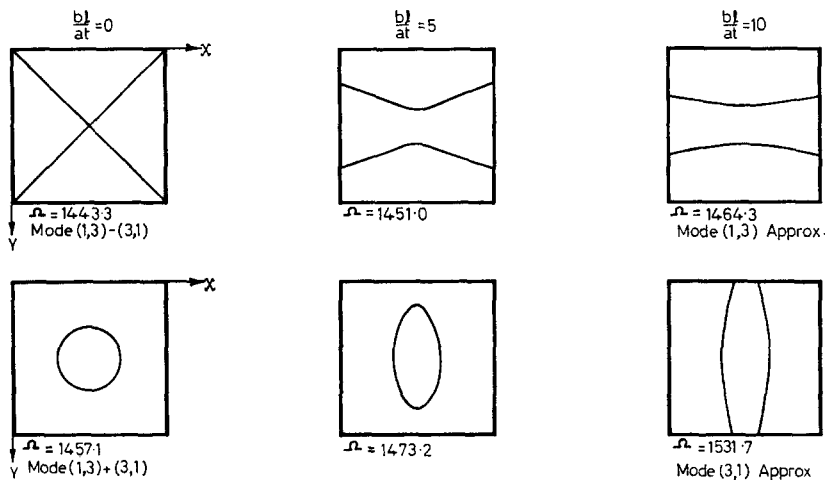


FIG. 7. Effect of curvature on the nodal patterns of clamped square curved panels, $b/t = 100$.

For example, in Figs. 2-6, the cross-over points between the modes with the lowest natural frequency are all between modes with different symmetry or anti-symmetry in the y direction. Hence, at the cross-over points, two modes exist with equal natural frequencies. Similarly it might be expected that a clamped, flat, square plate has a $(1, 3)$ and a $(3, 1)$ normal mode with equal natural frequencies. These modes have the same symmetry in the x direction and in the y direction and they are replaced by the two normal modes, shown in Fig. 7, which have different natural frequencies.

CONCLUSIONS

Simply supported panels

The solutions for simply supported panels given by equations (1), (2) and (3) are exact in so far as they are the exact solutions of Flügge's equations of motion. Shear deflexion and rotatory inertia were neglected in the formulation of the equations of motion and thus the solutions are only valid when these effects are negligible, i.e. for half-wavelengths in both the x and y directions greater than 10 thicknesses.

Equations (4) and (5) are more convenient to use than equations (2) and (3) but they should be used with discretion. The terms which are omitted from equation (4) to give equation (5), when Donnell's approximation is used, should be included if they are significant. An assessment of the effect of neglecting in-plane inertia may be made by calculating the amplitude ratios, A/C and B/C . A more accurate value of Δ is given by dividing the value given by equations (4) or (5) by $[1 + (A/C)^2 + (B/C)^2]$. This improved value of Δ is still greater than the value which would be obtained from the equations which include in-plane inertia.

Clamped panels

Again shear deflexion and rotatory inertia have been neglected in the analyses and the solutions given above should not be used if the ratios of the half-wavelengths to thickness are less than 10. In addition, Sewall's equation (12) is valid only if either bl/at or $b^2/at < 20$ and $b/a < 1$.

The double-power series method gives accurate results if a sufficient number of terms is used in the power series representing the displacements. The results plotted in Figs. 2-6 are for panels with $b/a = 0.1$ but they may be used for panels with $b/a \leq 1.0$.

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